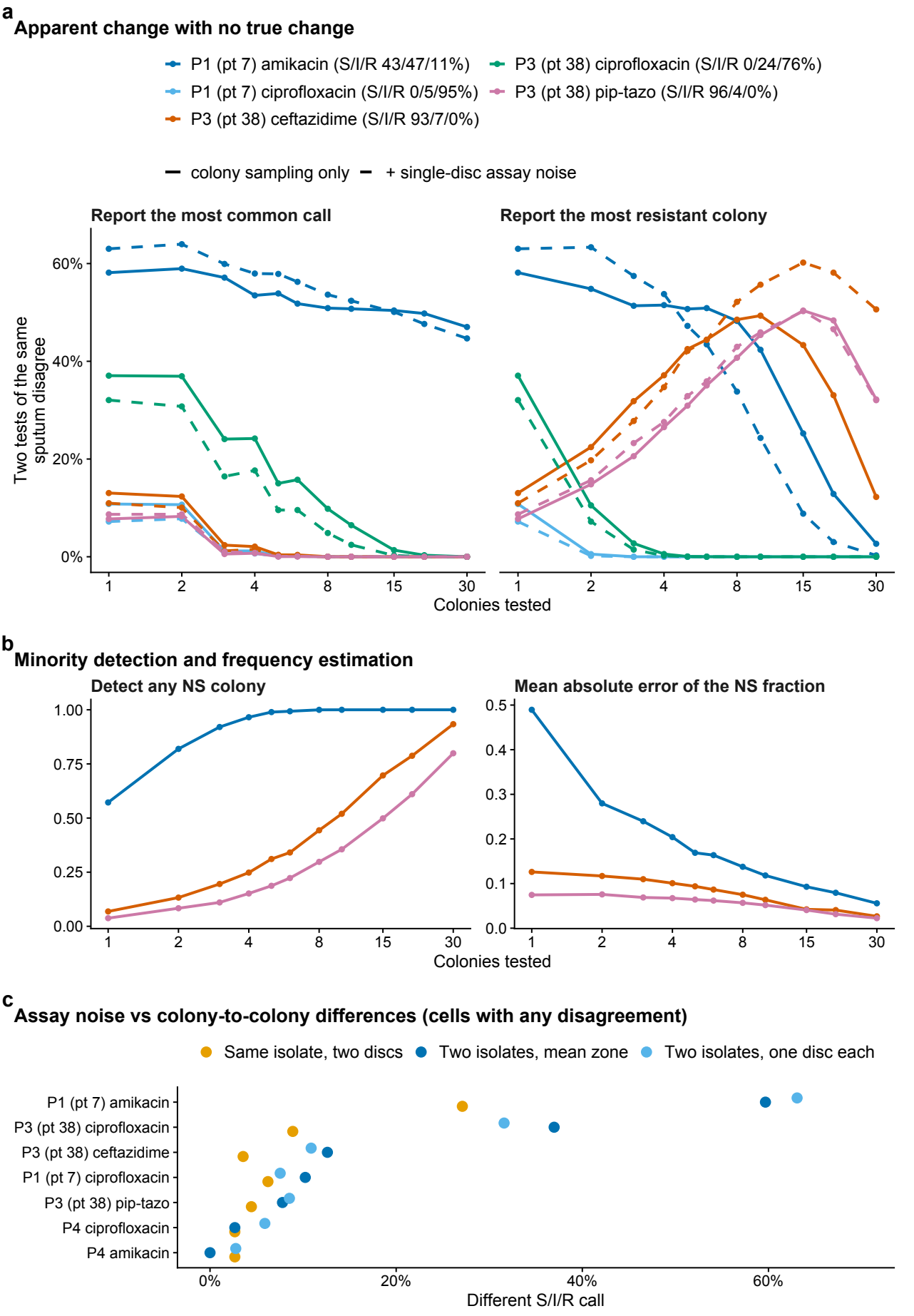

a
Apparent change with no true change
P1 (pt 7) amikacin (S/I/R 43/47/11%)
P1 (pt 7) ciprofloxacin (S/I/R 0/5/95%)
P3 (pt 38) ceftazidime (S/I/R 93/7/0%)
P3 (pt 38) ciprofloxacin (S/I/R 0/24/76%)
P3 (pt 38) pip-tazo (S/I/R 96/4/0%)
colony sampling only
+ single-disc assay noise
Report the most common call
Report the most resistant colony
Two tests of the same sputum disagree
60%
40%
20%
0%
1
2
4
8
15
30
Colonies tested
b
Minority detection and frequency estimation
Detect any NS colony
Mean absolute error of the NS fraction
1.00
0.75
0.50
0.25
0.00
0.5
0.4
0.3
0.2
0.1
0.0
Colonies tested
c
Assay noise vs colony-to-colony differences (cells with any disagreement)
Same isolate, two discs
Two isolates, mean zone
Two isolates, one disc each
P1 (pt 7) amikacin
P3 (pt 38) ciprofloxacin
P3 (pt 38) ceftazidime
P1 (pt 7) ciprofloxacin
P3 (pt 38) pip-tazo
P4 ciprofloxacin
P4 amikacin
0%
20%
40%
60%
Different S/I/R call